SokoSense Fix Plan

Sep 26, 2026 · @Sophie

What this is: A practical roadmap for fixing the hiccups we found in the findings report, plus the data, backend, and frontend gaps still open. Written so the team can read it without digging through code.

Companion doc: FINDINGS_REPORT.md (what we found) · this file (how we fix it)

Where we stand today

Already fixed and merged

- USSD replies were too long for the 182-character limit (SMS allows 320) — diagnosed, fixed, and tested in English and Swahili for all three services.
- SMS looked like it was working but never actually called Africa's Talking to send — fixed and verified in the sandbox.
- Loan advice SMS now shows the top 3 lenders with honest, lender-specific reasons (not the same reason repeated). USSD Loan Check also sends a fuller SMS follow-up.

Still open

Everything below — including the "farmer history / decision for farmers" work that was marked remaining/blocked.

The one thing that changes the order of work

Under the hood, the **market price pipeline is currently broken** (a bad merge left broken code). When prices fail, market and timing advice go quiet or give weak answers. So we fix **data first**, then engines, then farmer history, then the website, then the Admin view. Polishing the map before prices work would waste time.

The big idea in one sentence

Get trustworthy prices into the system → make SMS/USSD advice correct → start logging what farmers asked and what we recommended → show that live activity in the Admin dashboard.

```
flowchart TD
  Wave0[Wave 0: Unblock] --> Wave1[Wave 1: Fix market data]
  Wave1 --> Wave2[Wave 2: Fix SMS and USSD advice]
  Wave2 --> Wave3[Wave 3: Farmer decision history]
  Wave3 --> Wave4[Wave 4: Fix the website and loan dashboard]
  Wave3 --> Wave5[Wave 5: Admin live activity]
  Wave0 --> Creds[Secure Neo4j and API keys]
```

Wave 0 — Unblock (same day, under an hour)

Quick safety and “won’t even start” fixes before the real work.

What	Why it matters	In plain English
Neo4j password	Live database password was left in a public example file	Change the password, put fake placeholders in the example file
Agent import	One missing import can crash the AI agent on load	Commit the small fix already sitting in the working tree
Groq + Neo4j keys	Advisory and agent features need valid keys locally	Make sure real keys live only in private `.env`, never in git

Done when: App starts cleanly; secrets are not public; Groq and Neo4j work on a local smoke test.

Wave 1 — Fix market data (foundation)

Goal: Every one of our seven focus markets (Nairobi, Nakuru, Eldoret, Kisumu, Mombasa, Kitale, Nyeri) has usable prices for our six crops. If a market has no fresh price, we suggest the nearest one and say so honestly.

What’s wrong today

- The code that reads prices from our SQLite database is half-broken after a merge.
- The automatic KAMIS refresh only pulls a tiny slice of data, so some markets (e.g. Nyeri) are empty or nearly empty.
- County names don’t always match (hyphen vs space), so Eldoret/Kitale can be missed even when data exists.
- When a market has no price, farmers often get a dead end instead of a helpful alternative.

What we will do

1. Repair the price-reading code so market and timing engines use the database again.
2. Pull more pages of KAMIS data so all seven markets get baseline prices.
3. Treat “Uasin-Gishu” and “Uasin Gishu” as the same place.
4. If Meru (or similar) has no price, recommend the nearest tracked market and tell the farmer.
5. Make the nightly/3-hourly sync job always update the right branch, and fail loudly if it inserts zero rows.

Done when: Asking for maize prices returns real numbers for all seven markets (or a clear “nearest market” note). Market and timing no longer fail because the pipeline can’t load.

Why this comes first: SMS, USSD, the map, and demos all depend on this. Everything else is guessing until this works.

Wave 2 — Fix advice on SMS and USSD

Goal: Farmers get the right market, the right timing for *their* location, and loan maths that doesn’t confuse “amount borrowed” with “interest rate.”

Priority	Problem	Fix (plain English)
Must do	“No market found” dead end	Suggest the nearest/best market with a real price
Must do	“When to sell” on USSD always uses Nairobi	Ask where the farmer is (same as the market-price menu)
Must do	Loan SMS can treat 5000 as the interest rate	Parse amount vs rate correctly (look for `%` / “at 5%”)
Should do	Some places still use old 13% CBK rate; live rate is 8.75%	One shared 8.75% number everywhere
Should do	Duplicate dead code for parsing prices	Delete the unused copy so we don’t edit the wrong one
Should do	Unknown market silently becomes Nairobi	Either recognise more markets or say “showing nearest: …”
Nice to have	Billing only counts loans	Also record market and timing queries for the revenue demo

Done when: USSD timing for Eldoret uses Eldoret; `LOAN 5000 … at 5%` is read correctly; unknown markets get a nearest-market message; API loan checks show 8.75% CBK.

Wave 3 — Farmer decision history

Goal: Every SMS or USSD advice we give is logged (privacy-safe) so we can show partners a real decision trail — and so the Admin dashboard stops looking empty.

This was marked **remaining / blocked**. This plan unblocks it with a **small, useful first version** — not a full credit-score product yet.

What we will build

A simple log of decisions:

- Who (phone number stored as a hash — not plain text)
- How they asked (SMS or USSD)
- What they asked about (market / timing / loan, crop, place)
- What we replied
- When, and how long it took

That same log powers:

1. The Admin live activity view (built in Wave 5; today it always shows nothing)
2. An optional export partners can pull

What we are *not* doing in this wave

Full “financial identity” or automated credit scoring. That becomes Phase 2 once we have real history to score against.

Done when: One test SMS creates a row; opening Admin logs shows that row.

Wave 4 — Fix the website and loan dashboard

Goal: The web UI matches reality — no fake huge profits from missing prices, demos that don’t point at empty markets, and a loan page using the real CBK rate. The SACCO page is removed.

Problem	Fix (plain English)
Map treats missing price as zero and invents huge “arbitrage”	Ignore markets with no real price before doing maths
Landing and timing pages default to Nakuru, which may be empty	On load, pick a market that actually has a price
Loans page still shows 13% CBK in places	Use the rate the backend returns (8.75%)

Problem	Fix (plain English)
SACCO page is all mock data	Remove it completely: the page, its route, its nav links, and its mock data

Important: Do not start this wave's map polish until Wave 1 is done. Otherwise you "fix" the UI against broken or zero prices.

Done when: Map numbers look sane; demos use active markets; loans UI matches 8.75%; the SACCO page and every link to it are gone.

Wave 5 — Admin showing live activity

Goal: The Admin dashboard shows what is actually happening: which farmers asked what, on which channel, and what we replied.

Problem	Fix (plain English)
Admin logs page always shows nothing	Read from the farmer decision log built in Wave 3 (`/api/logs`)
No view of live activity	Show a feed of recent queries plus simple counts by channel (SMS/USSD), crop, and market

Important: This wave needs the Wave 3 log. Until it exists, build the page against sample data in the same shape.

Done when: Sending one test SMS makes it appear in the Admin live activity view.

How we'll ship the work (small PRs)

1. **Data PR** — Wave 1 (repair pipeline + refill the price database)
2. **Engines PR** — Wave 2 (SMS/USSD correctness)
3. **Farmer log PR** — Wave 3 (history + `/api/logs`)
4. **Frontend PR** — Wave 4 (map, defaults, loans page, remove SACCO page)
5. **Admin PR** — Wave 5 (live activity view)

Who does what

Each person owns their waves end to end. Everyone starts on day one: only the map work and the price-dependent advice wait for Job's Wave 1.

Person	Owns	Start now	Waits on Wave 1	Done when
Sophie	Wave 0 (today) + Wave 4 Frontend PR (loan dashboard)	Rotate Neo4j password, clean example file, commit agent import, keys into `.env`; loans page reads the rate from the backend; landing/timing pick a market that has a price; remove the SACCO page (page, route, nav links, mock data)	Map fix (ignore markets with no price)	App starts cleanly with no public secrets; loans page shows 8.75%; SACCO page gone; map numbers look sane
Job	Wave 1 Data PR (critical path)	Repair the SQLite price-reading code; pull more KAMIS pages; normalise county names (Uasin-Gishu); build one shared `nearest_market()` function; sync job fails loudly on zero rows	—	Real maize prices for all 7 markets, or a clear nearest-market note
Lucy	Wave 2 Engines PR + SMS/USSD demo	Loan parsing (amount vs rate); one shared 8.75% CBK constant; delete duplicate parsing code; USSD “when to sell” asks for location; billing records market and timing queries	Wiring Job’s `nearest_market()` into the “no market found” and unknown-market replies	Eldoret timing uses Eldoret; `LOAN 5000 … at 5%` parses correctly; 8.75% everywhere

Person	Owns	Start now	Waits on Wave 1	Done when
Ian	Wave 3 Farmer log PR + Wave 5 Admin PR	Decision log table (hashed phone) + `/api/logs` + a `log_decision()` helper; then the Admin live activity view built on `/api/logs`	Nothing	One test SMS creates a row and shows up in the Admin live activity view

How we'll know each wave worked

- **Wave 1:** Refresh prices locally; check maize (and beans) across all seven markets; try Nakuru and Nyeri on market + timing APIs.
- **Wave 2:** USSD "when to sell" for Eldoret; SMS loan with amount and rate; unknown market gets a nearest note; CBK shows 8.75%.
- **Wave 3:** Send one SMS → see it in the database → `/api/logs` returns it.
- **Wave 4:** Map has no inflated profit from zeros; landing/timing use live markets; loans page shows 8.75%; SACCO page is gone.
- **Wave 5:** Send one SMS → it appears in the Admin live activity view.

Bottom line

We already fixed the worst SMS/USSD delivery bugs. The next win is **reliable market prices**, then **honest advice on the phone**, then **a farmer decision log** that makes the Admin dashboard real. Do it in that order and every later fix builds on something solid.

Demo

Lucy will demo SMS and USSD.